FURTHER MECHANICS / CENTRE OF MASS

Centre of Mass of Laminae by Integration

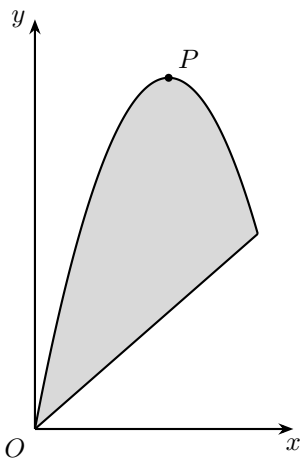
Questions by topic 10 questions · 115 marks

Instructions

Unless a question states otherwise, take $g = 9.8\,\mathrm{m\,s^{-2}}$ wherever a numerical value of g is required. Solutions also give final answers for $g = 10\,\mathrm{m\,s^{-2}}$ where these differ.

Questions

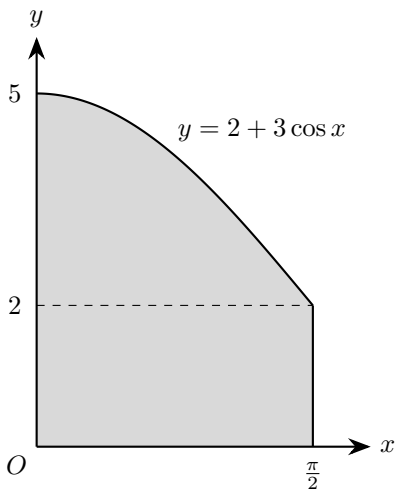
Question 1	*11 marks*	2
Question 2	*10 marks*	3
Question 3	*11 marks*	4
Question 4	*11 marks*	5
Question 5	*10 marks*	6
Question 6	*9 marks*	7
Question 7	*12 marks*	8
Question 8	*13 marks*	9
Question 9	*10 marks*	10
Question 10	*18 marks*	11



1. A uniform lamina occupies the region enclosed by the curve $y = 6x - x^2$ and the straight line $y = x$.
The unit of length on each axis is one centimetre.
The diagram above shows the lamina and the point $P(3, 9)$ at the maximum of the curve.

(a) Find the area of the lamina and use integration to find the coordinates of its centre of mass. Give your answers in exact form. **[8]**

(b) The lamina is freely suspended from P and hangs in equilibrium.
Find the acute angle between the vertical and the straight edge of the lamina, giving your answer to the nearest degree. **[3]**

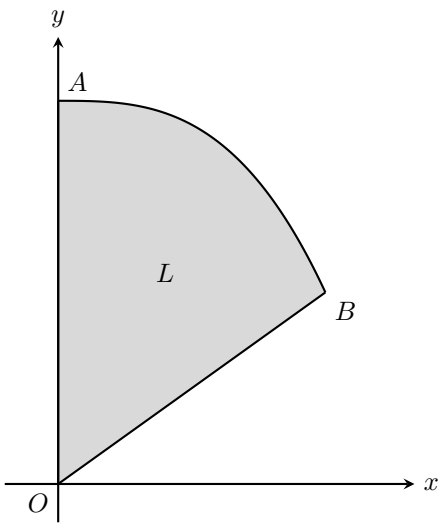


2. A uniform lamina L is cut to form a decorative panel.
It occupies the region bounded by the curve $y = 2 + 3\cos x$, the x-axis and the vertical lines $x = 0$ and $x = \frac{\pi}{2}$, as shown in the diagram above.
Its centre of mass is $(\bar{x}, \bar{y})$.

(a) Using integration, show that

$$\bar{x} = \frac{\pi^2 + 6\pi - 12}{4(\pi + 3)}$$ **[5]**

(b) Find the exact value of $\bar{y}$. Hence show, without using decimal approximations, that the centre of mass lies above the line $y = 2$. **[5]**



3. The diagram above shows a region bounded by the y-axis, the straight line $y = x$ and the curve

$$y = 6 - \frac{x^3}{9}$$

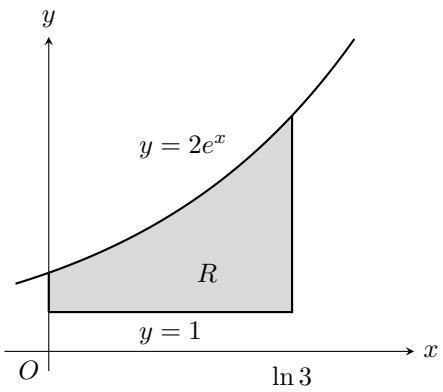
The curve meets the y-axis at $A = (0, 6)$ and meets the line $y = x$ at $B = (3, 3)$.
The origin is O.

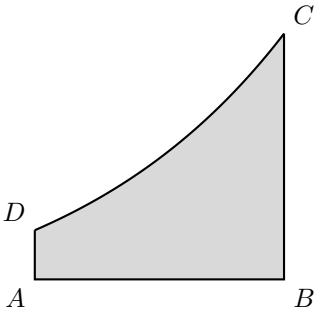
A uniform lamina L occupies the shaded region OAB.
The unit of length on each axis is one centimetre, and its centre of mass has coordinates $(\bar{x}, \bar{y})$.

(a) Use integration to find the area of L and show that

$$\bar{y} = \frac{118}{35}$$ **[5]**

(b) The lamina is freely suspended from B and hangs in equilibrium. Find the acute angle between the straight edge OB and the downward vertical, giving your answer in degrees to one decimal place. **[6]**





4. The shaded region R is bounded by the curve $y = 2e^x$, the horizontal line $y = 1$, the y-axis and the line $x = \ln 3$.
The unit of length on each axis is one metre.

A uniform lamina occupies R, and its centre of mass has coordinates $(\bar{x}, \bar{y})$.

(a) Use integration to find the area of R and show that

$$\bar{y} = \frac{16 - \ln 3}{8 - 2\ln 3}$$ **[5]**

The second diagram above shows the same lamina, with vertices $A = (0, 1)$, $B = (\ln 3, 1)$, $C = (\ln 3, 6)$ and $D = (0, 2)$.

The lamina is freely suspended from D and hangs in equilibrium.
Let θ be the angle between the directed segment from D to A and the downward vertical, where $0° \leq \theta \leq 180°$.

(b) Use integration to find $\bar{x}$ in exact form. Hence find θ, giving your answer to one decimal place. **[6]**

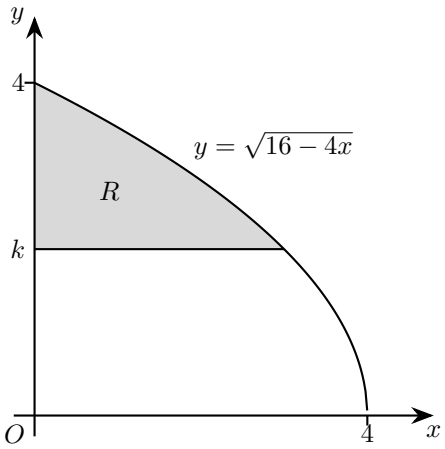
5. A uniform lamina occupies the region bounded by the curve

$$y = \frac{k}{x+1}$$

the x-axis and the vertical lines $x = 0$ and $x = 3$, where k is a positive constant.

(i) Find the exact x-coordinate of the centre of mass of the lamina. **[5]**

(ii) The centre of mass lies on the line $y = x$.
Find the exact value of k and hence the exact area of the lamina. **[5]**



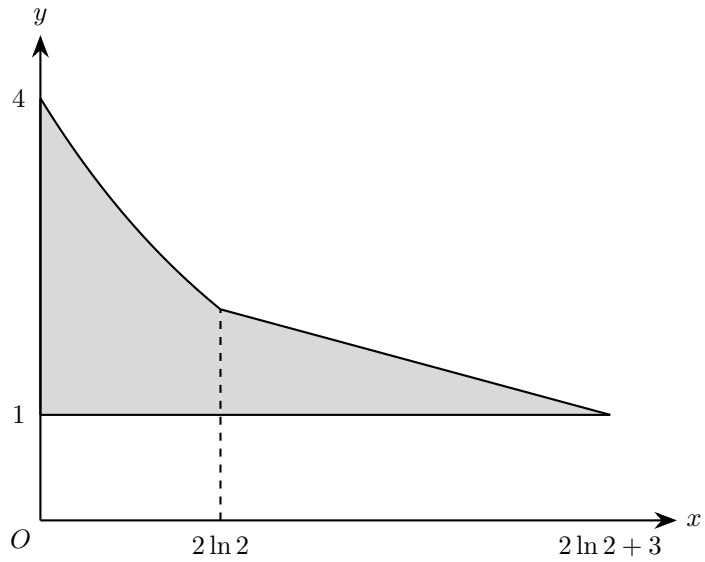
6. The shaded region R is bounded by the curve $y = \sqrt{16 - 4x}$, the y-axis and the horizontal line $y = k$, where $0 < k < 4$.
A uniform lamina occupies R.

(a) Using integration, determine the y-coordinate of the centre of mass of the lamina in terms of k. Simplify your expression fully. **[6]**

(b) The centre of mass lies on the line $y = 3$. Find the exact value of k, explaining why any other root of your equation must be rejected. **[3]**

7. A uniform lamina occupies the region between the curve $y = 4e^{-x/2}$ and the line $y = 1$, for $0 \leq x \leq 2\ln 2$.

(i) Using integration, find the exact area of this lamina and the exact y-coordinate of its centre of mass. **[5]**

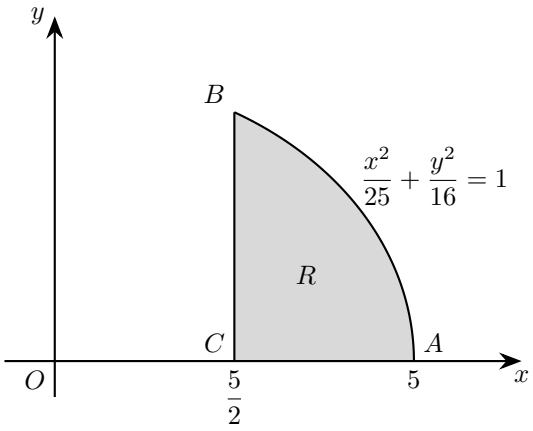


A triangular piece of lamina with the same uniform mass per unit area is attached to its right-hand edge. The resulting lamina is shown in the diagram above.
Its lower boundary is $y = 1$ and its upper boundary is $y = f(x)$, where

$$f(x) = \begin{cases} 4e^{-x/2}, & 0 \leq x \leq 2\ln 2 \\ 2 - \dfrac{x - 2\ln 2}{3}, & 2\ln 2 \leq x \leq 2\ln 2 + 3 \end{cases}$$

Its left-hand boundary lies on the y-axis.

(ii) Using your area from part (i), find the exact x-coordinate of the centre of mass of the resulting lamina. **[7]**



8. A uniform lamina occupies the shaded region R, bounded by the upper arc of the ellipse

$$\frac{x^2}{25}+\frac{y^2}{16}=1$$

the line $x=\frac{5}{2}$ and the x-axis.

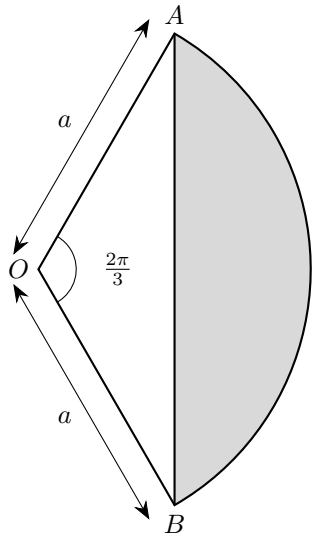
The points C and A have coordinates $\left(\frac{5}{2},0\right)$ and $(5,0)$ respectively.
The point B is the upper intersection of the ellipse with $x=\frac{5}{2}$.
The unit of length on each axis is one metre.

(a) Using the substitution $x=5\sin t$, show by integration that the area of R is

$$\frac{5}{6}\left(4\pi-3\sqrt{3}\right)\text{ m}^2$$ **[4]**

(b) Use integration to find the exact coordinates $(\bar{x},\bar{y})$ of the centre of mass of the lamina. **[6]**

(c) The lamina is freely suspended from A and hangs in equilibrium. Find the acute angle between AC and the downward vertical, giving your answer in degrees to one decimal place. **[3]**



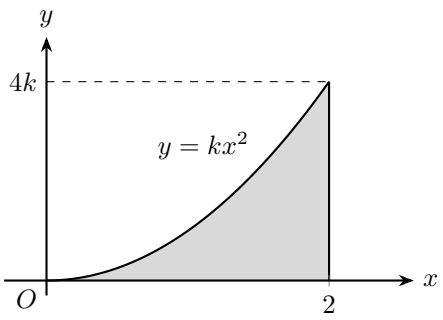
9. A thin uniform metal plate has the shape of the shaded circular segment in the diagram above.
It is bounded by the chord AB and the minor arc AB of a circle with centre O and radius a, where $a > 0$.
The angle AOB is $\frac{2\pi}{3}$.

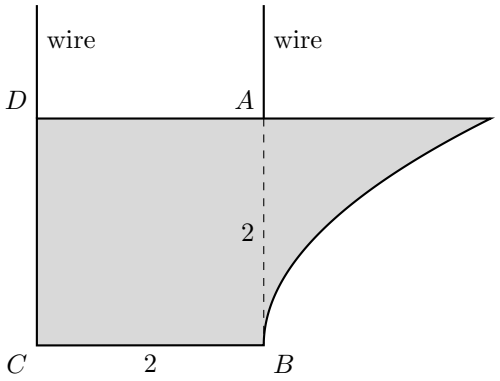
The plate is modelled as a uniform lamina with constant mass per unit area σ, where $\sigma > 0$.

Take O as the origin and the positive x-axis along the bisector of angle AOB, directed towards the minor arc.
Thus the chord AB lies on $x = \frac{a}{2}$.

(a) Using integration, find the mass of the plate in terms of σ and a. Give your answer in exact form. **[5]**

(b) Using integration to calculate the first moment of the plate about the y-axis, find the exact distance of its centre of mass from O. Explain why the centre of mass lies on the x-axis. **[5]**





10. A uniform lamina occupies the region bounded by the curve $y = kx^2$, the x-axis and the line $x = 2$, where $k > 0$, as shown in the first diagram above.

(i) Use integration to find both coordinates of the centre of mass of this lamina in terms of k. **[8]**

A sign is made from a uniform sheet by joining a square $ABCD$ of side length 2 to a curved piece having the shape of the lamina in part (i).
The sheet has the same thickness and density throughout.

For the sign, take C as the origin, with CB along the positive x-axis and CD along the positive y-axis.
Thus $A = (2, 2)$, $B = (2, 0)$ and $D = (0, 2)$.
The curved piece is obtained by reflecting the lamina in part (i) in the line $y = x$ and then translating it 2 units to the right.
Its curved edge therefore has equation

$$x = 2 + ky^2, \qquad 0 \leq y \leq 2$$

The complete sign is shown in the second diagram above.

(ii) Find both coordinates of the centre of mass of the sign in terms of k. **[5]**

The sign is held in equilibrium in a vertical plane by two vertical wires attached at A and D.
The tension in the wire at A is twice the tension in the wire at D.

(iii) Find the exact value of k. **[5]**